

TagFex: Task-Agnostic Guided Feature Expansion for Class-Incremental Learning

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Background

- **Continual Learning/Incremental Learning:** The ability to learn new concepts while preserving the learned knowledge is desirable for the learning system.
- **Class-Incremental Learning (CIL):** Learn new classes sequentially without forgetting old ones.
- Main challenge: **Catastrophic forgetting, class imbalance** (few old samples vs. many new) and **stability-plasticity trade-off**.
- Existing approaches:
 - **Model Expansion:** parameter isolation for each task.(Expands the feature space for each task)
 - **Rehearsal based methods:** store few old samples (limited by memory).
 - **Regularization based methods:** constrain weights or features
 - **Generative replay:** generate pseudo-samples (high compute cost).

Background(Contd.)

- **Model Expansion based approaches:** train a new feature extractor for each new task while keeping the previously learned model intact during the training of the new task.
- **Limitation:** Relies on feature diversity extracted by the new model and hence leads to feature collision of the new tasks and the old ones (models trained on different tasks capture different features), which in turn creates classification bias towards the new class.
- To mitigate this, train an auxiliary classifier and save a small number of rehearsal samples from the previous tasks to distinguish current task samples from previous samples. Helps the new model to capture diverse features.
- The limited number of rehearsal samples makes the training of the auxiliary classifier imbalanced and suboptimal.

Example

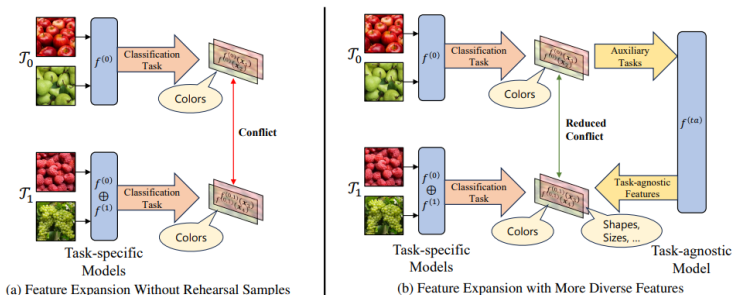


Figure 3. The illustration and mitigation of feature collision in expansion-based CIL. (a) Without the rehearsal samples, it is very difficult to distinguish two types of the fruits with the same color, because the features captured by just training on the newest task are insufficient to distinguish classes between two tasks. (b) With more diverse features, the model is able to further disengage from the spurious features, reducing the feature collision.

Figure: Example

Proposal

- This paper proposes to continually capture diverse features in a separate feature space, and then utilize it to help the incremental learning.
- **Task-specific Features:** In supervised classification within one task, the model only requires to capture minimal necessary features for the classification within the task. Such features are useful within a specific task, thus task-specific features.
- **Task-agnostic features:** There are plenty of features not discovered by these models containing more information , not necessarily related to the classification of the task, so task-agnostic features. If these features are available for subsequent tasks, the new task-specific model can refer to them for better feature learning.
- The proposed framework **TagFex** is composed of three parts:
 - The continual acquisition of the task-agnostic features
 - the adaptive feature aggregation with merge attention.
 - the task-agnostic guided knowledge transfer.

Overview

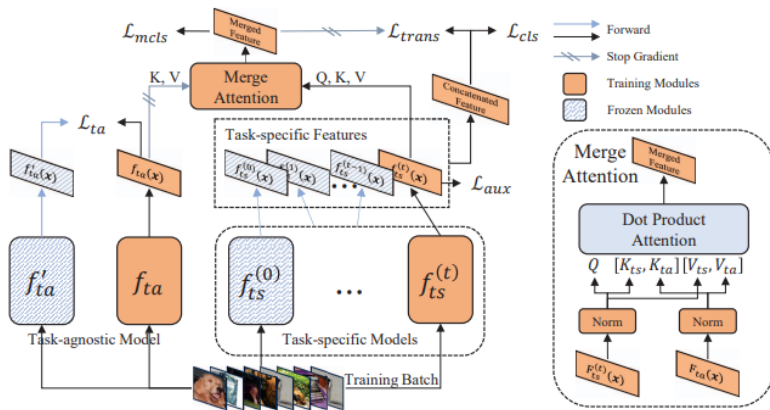


Figure: Overview of the proposed framework

TagFex Methodology

- The goal is simple: Make the model get better performance on all the seen task.
- Feature expansion of task-specific models(based on DER):
 - As shown in figure 2 a new model $f_{ts}^{(t)}$ is created at the beginning of each task.
 - The previously learned models $f_{ts}^{(0)} \sim f_{ts}^{(t-1)}$ are saved and freezed for feature extraction.
 - The classifier is also expanded and the trained parameter weights from the old classifier is inherited.
 - To predict the label of a sample, the concatenated feature from all of the models is used by the classifier. This is captured in $L_{cls}(\mathbf{x}, y)$ classification loss(in general cross-entropy loss).
 - An auxiliary classifier is trained based on the new feature to help the new model to capture diverse features.
 - The training objective of the auxiliary classifier is to distinguish classes within the current task and samples from the previous task. Hence, it outputs $|C_t| + 1$ logits, where the old classes are treated as one. The loss is again cross entropy $L_{aux}(\mathbf{x}, y)$.

TagFex Methodology(Cont,)

- Feature Diversity in Task-Specific Models
 - DER uses rehearsal samples and the auxiliary classifier to capture diverse features.
 - However, since the number of rehearsal samples are limited, the imbalanced training makes it suboptimal to capture more diverse features.
- To further improve the ability to capture more diverse features, this paper incorporates the task-agnostic features into the task-specific features.
- Continual Acquisition of Task-Agnostic Features:
 - Inspired by continual self-supervised learning(**CSSL**), which not only just extract the task-agnostic features continually but also preserves the old ones.
 - Follows similar method in **CaSSLe**, to continually capture more features in a predictive manner.
 - A separate model is trained to maintain the task-agnostic feature so that they remain unaffected by the classification objectives. L_{ta}
 - Save the trained task-agnostic model for the next task to preserve the predictive knowledge.

Task-agnostic loss

$$\mathcal{L}_{\text{ta}}^{(t=0)} = -\text{InfoNCE}(\{f_{\text{ta}}(\mathbf{x}_i)\}_{i=1}^B, \{f_{\text{ta}}(\mathbf{x}'_i)\}_{i=1}^B), \quad (3)$$

$$\begin{aligned} \mathcal{L}_{\text{ta}}^{(t>0)} = & -\text{InfoNCE}(\{f_{\text{ta}}(\mathbf{x}_i)\}_{i=1}^B, \{f_{\text{ta}}(\mathbf{x}'_i)\}_{i=1}^B) \quad (4) \\ & -\text{InfoNCE}(\{g(f_{\text{ta}}(\mathbf{x}_i))\}_{i=1}^B, \{f'_{\text{ta}}(\mathbf{x}_i)\}_{i=1}^B), \end{aligned} \quad (5)$$

where $\text{InfoNCE}(\{\mathbf{z}_i\}_{i=1}^B, \{\mathbf{z}'_i\}_{i=1}^B)$

$$= \frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}'_i)/\tau)}{\sum_{j=1, j \neq i}^B \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}'_j)/\tau)}, \quad (6)$$

Figure: $\mathcal{L}_{\text{ta}}$

- Using contrastive InfoNCE loss to maximize the lower bound of the mutual information between the sample and its representation.
- Adaptive Feature Aggregation by Merge Attention:
 - *Merged attention* is used to adaptively unify task-agnostic features with task-specific features in the incremental tasks.
 - In order to absorb useful features from task-agnostic features, the query comes from the task-specific representation, and the key and values comes from both the task-agnostic and task-specific(2).
 - Hence the merged feature contains the information learned from the current task and also the relevant information from the task-agnostic feature.
 - The merge classifier is trained to predict the label of the current task samples. $L_{mcls}(\mathbf{O}, y)$

Attention Maps

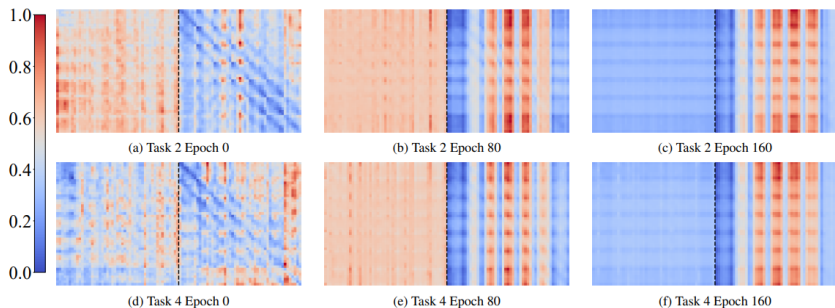


Figure: Attention Maps

TagFex Methodology(Contd.)

- Task-agnostic Guided knowledge Transferring:
 - Do not directly use the logits of the merge classifier for the final inference, since the distributional drift happens when the task-agnostic model is continually updated.
 - Have to transfer the diverse features from merged features to task-specific features. In the end, use the task-specific models with the classifier for inference.
 - Use **KL** divergence for knowledge transferring, between the logits of the merge classifier and the task-specific classifier. The loss is $L_{trans}(p_m, p_{ts})$
 - The overall loss we apply on the task-specific model is $L_{ts} = L_{cls} + L_{aux} + L_{trans}$.
 - The task-agnostic features are acquired continually by a separate model, and are first aggregated with the task-specific features and then transferred back into the task-specific features.
 - The overall loss of the framework during the training of each task is the the weighted sum of the three parts: $L = \lambda_{ta}L_{ta} + \lambda_{mcls}L_{mcls} + \lambda_{ts}L_{ts}$

Comparison: TagFex vs Previous Works

- Compared to previous expansion based methods, TagFex has significant performance improvements. Specifically, its adaptation of expansion strategy from **DER** along with task-agnostic model improved the performance by 3 – 4%.
- Also, the strategy to learn the task-agnostic features is adapted from **CaSSLe** through continual self-supervised learning(**CSSL**) but unlike it, TagFex does not require explicit task labels during training and does not rely on task boundaries.

Results

Table 1. Performance results. Bold font represents the best results in the scenario. TagFex-P is the method with pruning strategy.

Methods	CIFAR100				ImageNet100				ImageNet1000	
	10-10		50-10		10-10		50-10		100-100	
	Last	Avg	Last	Avg	Last	Avg	Last	Avg	Last	Avg
iCaRL [34]	49.52	64.64	50.56	60.08	50.98	67.11	53.69	62.56	40.47	57.55
BiC [50]	50.79	65.38	43.82	57.04	42.40	65.13	49.90	66.36	-	-
DyTox [10]	60.43	73.50	-	-	67.61	76.51	-	-	59.75	68.14
BEEF [44]	60.98	71.94	63.51	70.71	68.78	77.62	70.98	77.27	58.67	67.09
DER [51]	64.35	75.36	65.27	72.60	66.71	77.18	71.08	77.71	58.83	66.87
TagFex	68.23	78.45	70.33	75.87	70.84	79.27	75.54	80.64	61.45	68.32
TagFex-P	67.34	78.02	69.26	74.24	69.21	78.56	74.13	79.85	60.14	67.65

Figure: Performance Results

Ablation Study

Table 3. Ablation study on CIFAR100.

Task-agnostic Model w/ Merge Attention	Continual Task- agnostic Model	Knowledge Transferring	10-10		50-10	
			Last	Avg	Last	Avg
✓	✓		64.45	75.34	65.45	72.58
✓		✓	65.86	76.32	67.51	73.29
✓	✓	✓	68.23	78.45	70.33	75.87

Figure: Ablation Study to study the effectiveness of each part

Limitations

- Expansion modules add **extra parameters** → higher memory cost than pure regularization.
- Dominant new-class features may still bias decision boundaries.
- Requires careful balancing of **expansion vs. overfitting**.
- Theoretical analysis of long-term scalability still open.

Conclusion

- TagFex advances Class-Incremental Learning by:
 - Expanding feature space dynamically.
 - Combining predictive knowledge preservation (from CaSSLe) with improved plasticity.
 - Mitigating imbalance through richer old-class features.
- **Future Work:**
 - Better memory-efficient expansion mechanisms.
 - Deeper theoretical analysis of feature subspace growth.

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